Stem Cells: Evidence Before Judgement

LAUNCH · Science · 40 minutes

I can describe stem-cell functions and distinguish scientific evidence, ethical concern and uncertainty.

Name: ____________________________ Date: _______________

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
stem cell	unspecialised starting cell
differentiate	become specialised
ethical	about what ought to be done

First idea

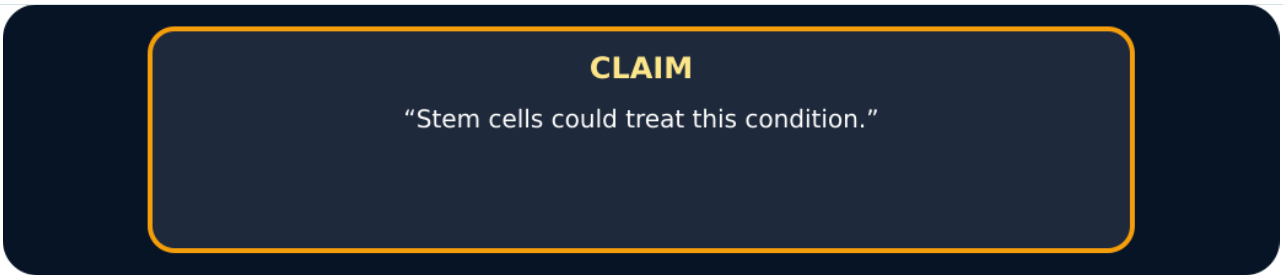
A headline says, 'Stem cells will cure every damaged organ.' What should a scientist do first?

A. Accept it because stem cells can differentiate.

B. Check the evidence, the cell type, the outcomes and the risks.

C. Reject every possible medical use of stem cells.

My first idea and reason:



Claim

We do · choose, explain, check

Which statement describes a stem-cell function rather than only a property?

A. It can divide and produce cells that differentiate, supporting growth or replacement.

B. It is called undifferentiated.

C. It cures every disease.

My choice: ______ Evidence or reason:

Evidence, ethics or uncertainty?

Place each statement by its main role. Then explain why one statement could matter to more than one column.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
SCIENTIFIC EVIDENCE	
ETHICAL CONCERN	
UNCERTAINTY	

Activity cards

1 · Observed division	2 · Consent matters
Cells in a controlled culture divided and some differentiated into a named cell type.	People disagree about when and how cells may be donated and used.

3 · Long-term outcome unknown
The study has not followed participants for long enough to measure later effects.

Which column contains a fact, which contains a value question and which names a gap in evidence?

Your lesson record

Build a stem-cell evidence glossary and classify three short statements before writing one cautious scientific claim.

Model helps us see

The three-column model keeps scientific findings, ethical concerns and uncertainty visible without pretending they are the same type of statement.

Model does not show

Real decisions involve more evidence, different stakeholders and changing medical knowledge; three cards cannot settle an ethical question.

Supported

Match stem cell, differentiate and ethical to icon-backed meanings; sort three statements.

Keep the word bank visible and use: This is __ because it says __.

Standard

Classify three statements and describe one stem-cell function before writing one cautious claim.

Use: Evidence shows __. It does not yet show __.

Stretch

Classify three statements, identify one overlap and explain why function alone cannot prove treatment success.

Name evidence, value and uncertainty separately; avoid adding Higher-tier content.

Evidence target

An accurate glossary, classified statements and one cautious claim that distinguishes possibility from evidence.

Exit, Audience and evidence • W11L1

Supported: Complete: a stem cell can divide and __.

Standard: Distinguish scientific evidence from an ethical concern in this context.

Stretch: Why does a plausible biological mechanism not prove that a treatment is safe and effective?

What I said, and what it changed

Next I will ______.

Adult witness note

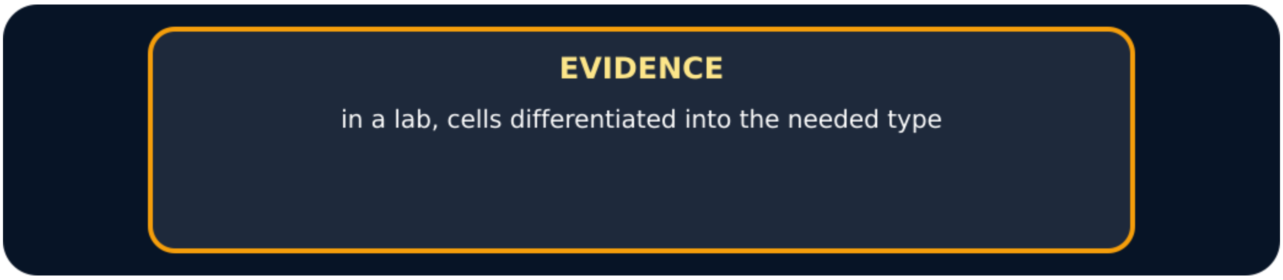
What was observed, and with what level of independence?

My evidence and explanation

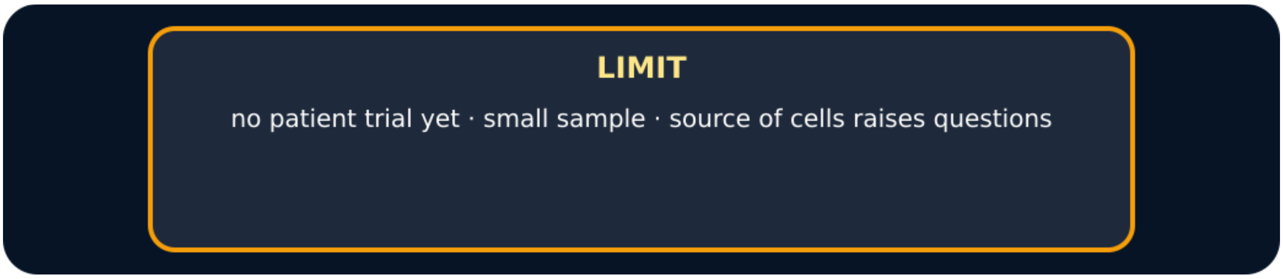
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Evidence



Limit

Exit • one next step

After the adult has listened, what will you keep, improve or check next?